

Chemistry
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Thinkers

Small
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NEET | JEE
Foundation

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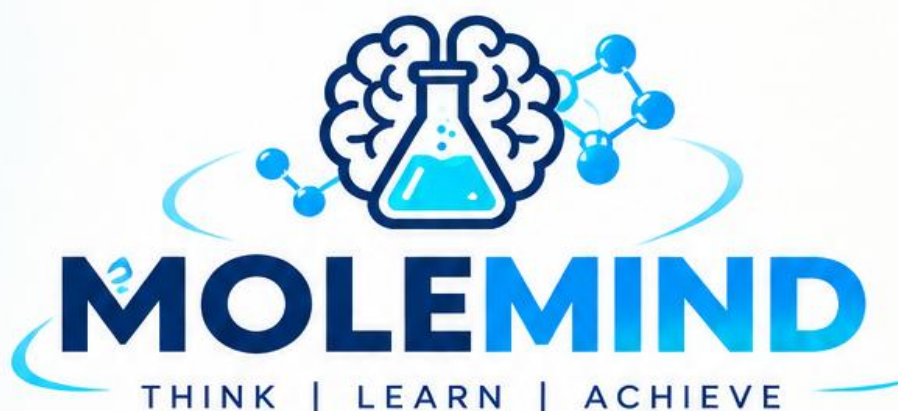
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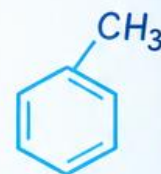
 Concepts
→ Practice
→ Success

Dr. Homi Bhabha Exam Practice Question Set



Acids, Bases and Salts Question Bank


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Organic
Chemistry
is not just a subject...
it's a way of thinking
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From Confusion to Clarity

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ACIDS, BASES AND SALTS

Basic Concepts

Q1. Which of the following is an acid?

- A) Sodium hydroxide
- B) Hydrochloric acid
- C) Calcium hydroxide
- D) Ammonium hydroxide

Q2. Which of the following is a base?

- A) HCl
- B) H_2SO_4
- C) NaOH
- D) CH_3COOH

Q3. Acids generally have a:

- A) Bitter taste
- B) Sweet taste
- C) Sour taste
- D) Salty taste

Q4. Bases generally have a:

- A) Sour taste
- B) Bitter taste
- C) Sweet taste
- D) Salty taste

Q5. Which substance is commonly present in lemon juice?

- A) Acetic acid
- B) Citric acid
- C) Hydrochloric acid
- D) Sulphuric acid

Q6. The acid present in vinegar is:

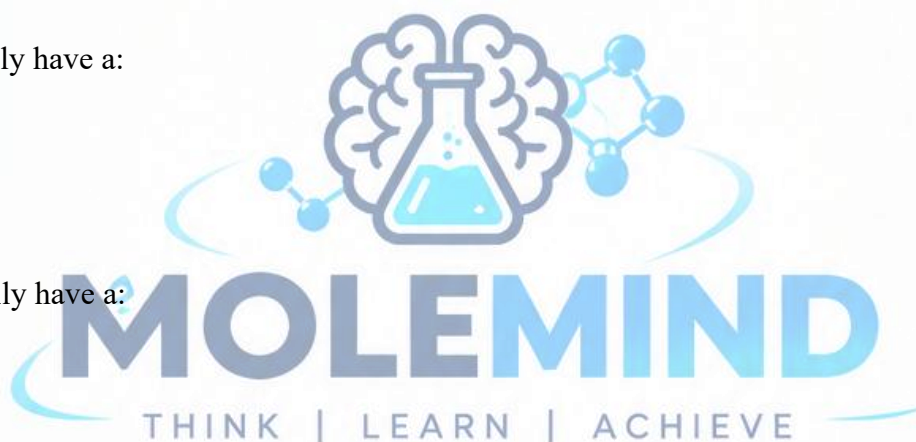
- A) Citric acid
- B) Acetic acid
- C) Lactic acid
- D) Formic acid

Q7. The acid present in curd is mainly:

- A) Lactic acid
- B) Citric acid
- C) Oxalic acid
- D) Acetic acid

Q8. Which acid is present in an ant sting?

- A) Citric acid
- B) Formic acid

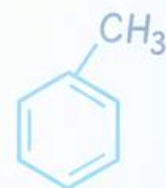


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- C) Lactic acid
D) Acetic acid

Q9. Which of the following is a natural indicator?

- A) Phenolphthalein
B) Methyl orange
C) Litmus
D) Sodium chloride

Q10. Litmus is obtained from:

- A) Algae
B) Lichens
C) Fungi only
D) Mosses

Q11. Blue litmus turns _____ in an acidic solution.

- A) Blue
B) Green
C) Red
D) Yellow

Q12. Red litmus turns _____ in a basic solution.

- A) Blue
B) Red
C) Yellow
D) Colourless

Q13. Phenolphthalein is generally _____ in an acidic solution.

- A) Pink
B) Red
C) Colourless
D) Blue

Q14. Phenolphthalein turns _____ in a basic solution.

- A) Pink
B) Yellow
C) Red
D) Green

Q15. Methyl orange turns _____ in an acidic solution.

- A) Yellow
B) Red
C) Blue
D) Green

Q16. Methyl orange is _____ in a basic solution.

- A) Red
B) Yellow
C) Pink
D) Blue

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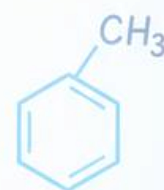
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A) Litmus
B) Phenolphthalein
C) Methyl orange
D) Sodium chloride

A) Acidic solution
B) Basic solution
C) Neutral solution only
D) Distilled water

A) Always produces a gas
B) Changes colour depending on the nature of solution
C) Always changes temperature
D) Forms a salt only

A) Soap solution
B) Lime water
C) Lemon juice
D) Baking soda solution

Q21. Acids react with many metals to produce:

A) Oxygen
B) Hydrogen
C) Nitrogen
D) Carbon dioxide

A) O_2
B) CO_2
C) H_2
D) Cl_2

A) ZnCl_2 and H_2
B) ZnO and H_2O
C) ZnCO_3 and H_2
D) ZnSO_4 and O_2

A) Zinc
B) Magnesium
C) Iron
D) Copper

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- B) Red
C) Yellow
D) Pink

Q34. Bases turn red litmus:

- A) Blue
B) Yellow
C) Green
D) Red

Q35. Which of the following is a basic solution?

- A) HCl solution
B) Lemon juice
C) NaOH solution
D) Vinegar

Q36. Which solution would feel soapy to touch?

- A) Dilute HCl
B) Sodium hydroxide solution
C) Lemon juice
D) Vinegar

Q37. Which of the following is NOT a typical property of acids?

- A) Sour taste
B) Turning blue litmus red
C) Bitter taste
D) Reaction with carbonates

Q38. Which of the following is NOT a typical property of bases?

- A) Bitter taste
B) Soapy feel
C) Turning red litmus blue
D) Sour taste

Q39. When an acid reacts with a metal oxide, the products are generally:

- A) Salt and water
B) Salt and hydrogen
C) Base and oxygen
D) Acid and carbon dioxide

Q40. Copper(II) oxide reacts with hydrochloric acid to form:

- A) Copper chloride and water
B) Copper sulphate and hydrogen
C) Copper carbonate and water
D) Copper hydroxide and chlorine

Neutralisation and Salts

Q41. The reaction between an acid and a base is called:

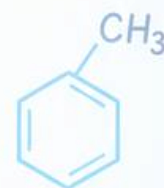
- A) Oxidation
B) Neutralisation

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- C) Reduction
D) Sublimation

Q42. The products of a typical neutralisation reaction are:

- A) Salt and water
B) Hydrogen and oxygen
C) Acid and metal
D) Carbon dioxide and water

Q43. $\text{HCl} + \text{NaOH} \rightarrow$

- A) $\text{NaCl} + \text{H}_2\text{O}$
B) $\text{Na}_2\text{CO}_3 + \text{H}_2$
C) $\text{NaOH} + \text{Cl}_2$
D) $\text{NaHCO}_3 + \text{O}_2$

Q44. In the reaction $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$, NaCl is a:

- A) Base
B) Acid
C) Salt
D) Indicator

Q45. Which salt is formed when hydrochloric acid reacts with sodium hydroxide?

- A) Na_2SO_4
B) NaCl
C) Na_2CO_3
D) KCl

Q46. Which salt is formed when sulphuric acid reacts completely with sodium hydroxide?

- A) NaCl
B) Na_2SO_4
C) Na_2CO_3
D) NaNO_3

Q47. Neutralisation is useful in treating:

- A) Acidic soil
B) Only pure water
C) Oxygen deficiency
D) Rusting only

Q48. Acidic soil can be treated by adding:

- A) Organic acids
B) Quicklime or slaked lime
C) Vinegar
D) Hydrochloric acid

Q49. Basic soil can be treated by adding:

- A) Organic matter/compost
B) Sodium hydroxide
C) Quicklime
D) Baking soda

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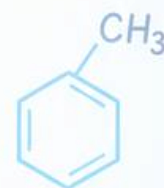
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Q50. The sting of an ant contains:

- A) Formic acid
- B) Acetic acid
- C) Citric acid
- D) Hydrochloric acid

Q51. The effect of an ant sting can be relieved by applying a mild:

- A) Acid
- B) Base
- C) Salt only
- D) Indicator

Q52. Which substance can help neutralise the acid from an ant sting?

- A) Baking soda
- B) Lemon juice
- C) Vinegar
- D) HCl

Q53. Factory wastes are often treated with appropriate chemicals before disposal to:

- A) Increase their acidity
- B) Neutralise harmful acids or bases
- C) Increase their temperature
- D) Produce metals

Q54. Which of the following represents neutralisation?

- A) Acid + base $\rightarrow$ salt + water
- B) Acid + metal $\rightarrow$ hydrogen only
- C) Salt $\rightarrow$ acid + base
- D) Base + water $\rightarrow$ acid

Q55. If equal quantities of a strong acid and strong base of equal concentration are mixed, the resulting solution is approximately:

- A) Strongly acidic
- B) Strongly basic
- C) Neutral
- D) Always salty and acidic

pH and Strength of Acids/Bases

Q56. The pH scale is used to measure:

- A) Temperature
- B) Acidity/basicity of a solution
- C) Density
- D) Mass

Q57. The pH scale generally ranges from:

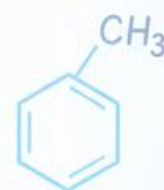
- A) 0 to 7 only
- B) 1 to 10
- C) 0 to 14
- D) 7 to 14 only

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Q58. A solution with pH 7 is generally:

- A) Acidic
- B) Basic
- C) Neutral
- D) Strongly acidic

Q59. A solution with pH less than 7 is:

- A) Acidic
- B) Basic
- C) Neutral
- D) Always concentrated

Q60. A solution with pH greater than 7 is:

- A) Acidic
- B) Basic
- C) Neutral
- D) Always dilute

Q61. Which solution is more acidic?

- A) pH 2
- B) pH 5
- C) pH 7
- D) pH 9

Q62. Which solution is more basic?

- A) pH 3
- B) pH 6
- C) pH 8
- D) pH 12

Q63. Among the following, which has the highest concentration of H^+ ions?

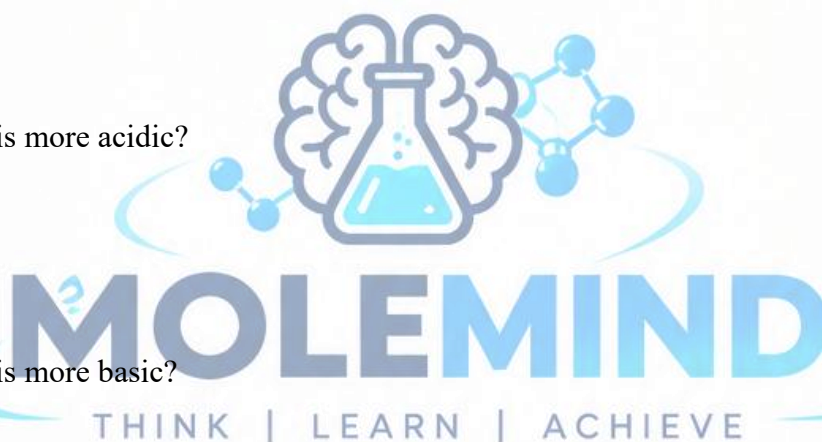
- A) pH 2
- B) pH 4
- C) pH 6
- D) pH 7

Q64. As the concentration of H^+ ions increases, pH generally:

- A) Increases
- B) Decreases
- C) Remains exactly the same
- D) Becomes 14

Q65. A solution of pH 3 is how many times more acidic in terms of H^+ concentration than a solution of pH 5?

- A) 2 times
- B) 10 times
- C) 100 times
- D) 1000 times

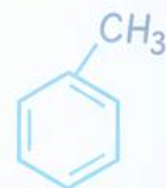


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A) 100 times greater
B) 10 times greater
C) 100 times smaller
D) Equal

A) pH 2
B) pH 5
C) pH 7
D) pH 10

A) 8
B) 10
C) 12
D) 14

A) 1
B) 3
C) 5
D) 6

A) 7.0
B) 6.5
C) 5.5
D) 2.0

A) 2.0–3.0
B) 4.0–5.0
C) 7.35–7.45
D) 10–12

A) Antacids
B) Strong acids
C) Sodium chloride only
D) Indicators

A) Acidic
B) Basic
C) Neutral gases
D) Strong oxidising agents

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Q74. Which of the following is commonly used as an antacid?

- A) Magnesium hydroxide
- B) Hydrochloric acid
- C) Sulphuric acid
- D) Nitric acid

Q75. Tooth decay is promoted when the pH of the mouth becomes:

- A) Higher than 10
- B) Lower than about 5.5
- C) Exactly 7
- D) Higher than 14

Q76. Toothpaste is generally:

- A) Acidic
- B) Basic
- C) Strongly acidic
- D) Neutral gas

Q77. The basic nature of toothpaste helps to:

- A) Increase acidity in the mouth
- B) Neutralise excess acids
- C) Produce carbon dioxide
- D) Dissolve enamel deliberately

Q78. Plants generally grow best in soil having:

- A) Any pH whatsoever
- B) A suitable pH range specific to the plant
- C) Only pH 14
- D) Only pH 1

Q79. Acid rain can be harmful to aquatic life because it can:

- A) Increase the pH of water greatly
- B) Lower the pH of water
- C) Remove all oxygen instantly
- D) Convert water into salt

Q80. The pH of rainwater can decrease due to atmospheric:

- A) Oxygen only
- B) Sulphur and nitrogen oxides
- C) Helium
- D) Hydrogen only

Salts & Important Compounds

Q81. Common salt is chemically known as:

- A) Sodium carbonate
- B) Sodium chloride
- C) Sodium hydroxide
- D) Calcium chloride

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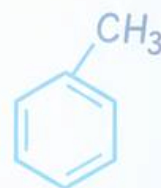
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Q82. The chemical formula of common salt is:

- A) NaOH
- B) NaCl
- C) Na_2CO_3
- D) NaHCO_3

Q83. Baking soda is:

- A) Sodium chloride
- B) Sodium hydrogen carbonate
- C) Sodium carbonate
- D) Calcium carbonate

Q84. The formula of baking soda is:

- A) Na_2CO_3
- B) NaHCO_3
- C) NaOH
- D) NaCl

Q85. Baking soda reacts with acids to produce:

- A) CO_2
- B) O_2
- C) H_2
- D) N_2

Q86. Which compound is commonly used in baking powder?

- A) Sodium hydrogen carbonate
- B) Sodium chloride only
- C) Calcium hydroxide only
- D) Hydrochloric acid

Q87. Washing soda is chemically:

- A) Sodium chloride
- B) Sodium carbonate decahydrate
- C) Sodium hydrogen carbonate
- D) Calcium sulphate

Q88. The formula of washing soda is:

- A) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- B) NaHCO_3
- C) NaCl
- D) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$

Q89. Washing soda is used in:

- A) Removing permanent hardness of water
- B) Making oxygen
- C) Neutralising only strong acids in laboratories
- D) Producing hydrogen gas

Q90. Bleaching powder is commonly represented as:

- A) CaOCl_2

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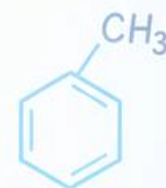
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- B) NaCl
C) Na_2CO_3
D) CaCO_3

Q91. Bleaching powder is commonly used for:

- A) Disinfecting drinking water
B) Increasing water hardness
C) Producing table salt
D) Making acids stronger

Q92. Plaster of Paris has the formula:

- A) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
B) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
C) CaCO_3
D) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

Q93. Gypsum has the formula:

- A) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
B) $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$
C) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
D) CaOCl_2

Q94. Plaster of Paris is obtained by heating:

- A) Sodium chloride
B) Gypsum
C) Baking soda
D) Limestone only

Q95. When water is added to Plaster of Paris, it changes into:

- A) Gypsum
B) Sodium chloride
C) Baking soda
D) Quicklime

Tricky & Application-Based MCQs

Q96. A solution turns blue litmus red and phenolphthalein remains colourless. The solution is most likely:

- A) Acidic
B) Basic
C) Neutral
D) Strongly alkaline

Q97. A solution turns red litmus blue and phenolphthalein pink. It is:

- A) Acidic
B) Basic
C) Neutral
D) A salt solution necessarily

Q98. Solution A has pH 2 and solution B has pH 4. Compared with B, solution A has H^+ ion concentration that is:

- A) 2 times greater

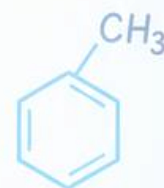
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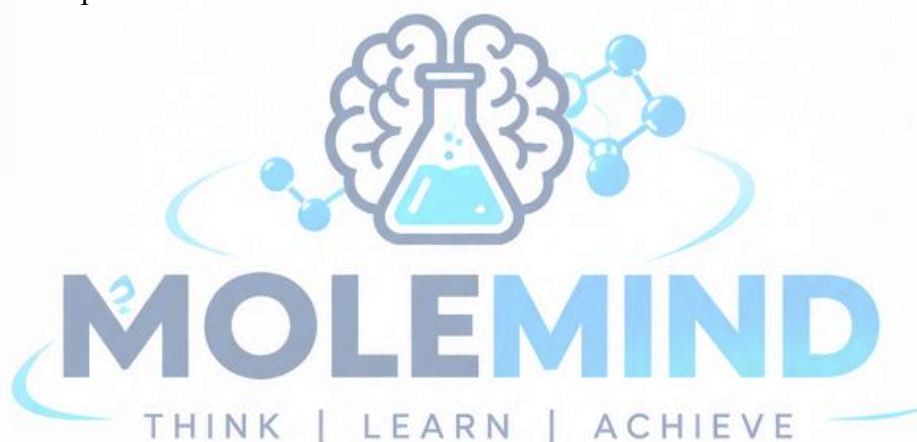
- B) 10 times greater
- C) 100 times greater
- D) 100 times smaller

Q99. A student adds dilute HCl to sodium hydrogen carbonate. Which observation is expected?

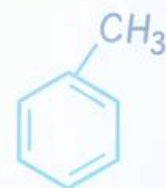
- A) Evolution of CO_2 gas
- B) Evolution of H_2 gas
- C) Formation of O_2 gas
- D) No reaction

Q100. A farmer finds that the soil is too acidic for a particular crop. Which action would be most appropriate?

- A) Add a suitable basic substance such as lime
- B) Add more hydrochloric acid
- C) Add concentrated sulphuric acid
- D) Add vinegar



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